IMO Training
Algebraic Combinatorics: Linear Algebra and Set Systems

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Introduction

Definition (Vector Space and Vector Subspace)**.** An $\mathbb{F}$-*vector space* is an (additive) abelian group V together with a function $\mathbb{F} \times V \to V$, written $(\lambda, \mathbf{v}) \mapsto \lambda\mathbf{v}$, such that

1. $\lambda(\mu\mathbf{v}) = \lambda\mu\mathbf{v}$ for all $\lambda, \mu \in \mathbb{F}$, $\mathbf{v} \in V$ (associativity)
2. $\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v}$ for all $\lambda \in \mathbb{F}$, $\mathbf{u}, \mathbf{v} \in V$ (distributivity in V)
3. $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}$ for all $\lambda, \mu \in \mathbb{F}$, $\mathbf{v} \in V$ (distributivity in $\mathbb{F}$)
4. $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in V$ (identity)

If $W \subset V$ and W itself is a vector space, then W is a *vector subspace* of V. We write $W \leq V$.

Definition (Span)**.** Let V be a vector space over $\mathbb{F}$ and $S \subseteq V$. The *span* of S is defined as

$$\langle S \rangle = \left\{ \sum_{i=1}^{n} \lambda_i \mathbf{s}_i : \lambda_i \in \mathbb{F}, \mathbf{s}_i \in S, n \geq 0 \right\}$$

This is the smallest subspace of V containing S. We say S *spans* V if $\langle S \rangle = V$.

Definition (Linear Independence)**.** Let V be a vector space over $\mathbb{F}$ and $S \subseteq V$. Then S is *linearly independent* if whenever

$$\sum_{i=1}^{n} \lambda_i \mathbf{s}_i = \mathbf{0} \text{ with } \lambda_i \in \mathbb{F}, \mathbf{s}_1, \mathbf{s}_2, \cdots, \mathbf{s}_n \in S \text{ distinct,}$$

we must have $\lambda_i = 0$ for all i. If S is not linearly independent, we say it is *linearly dependent*.

Definition (Basis and Dimension)**.** Let V be a vector space over $\mathbb{F}$ and $S \subseteq V$. Then S is a *basis* for V if S is linearly independent and spans V. Furthermore, if S is finite, the *dimension* of V is $\dim V = |S|$.

Definition (Image and kernel of matrix)**.** The *image* or *column space* $\mathrm{im}(\mathbf{A})$ of a matrix $\mathbf{A}$ is the span of its column vectors. The *kernel* or *null space* $\ker(\mathbf{A})$ of the matrix is the set of vectors $\mathbf{x}$ which satisfy $\mathbf{A}\mathbf{x} = \mathbf{0}$.

Theorem (Rank-nullity theorem)**.** The dimensions of the image and kernel of an $m \times n$ matrix $\mathbf{A}$ are related by the *rank-nullity theorem*: $\mathrm{rank}(\mathbf{A}) + \dim(\ker(\mathbf{A})) = n$, where $\mathrm{rank}(\mathbf{A})$ denotes the *rank* of the matrix or the dimension of the image $\dim(\mathrm{im}(\mathbf{A}))$.

Definition. For W a vector subspace of V, denote the *orthogonal complement* by $W^\perp = \{\mathbf{v} \in V \mid \mathbf{v} \cdot \mathbf{w} = 0 \ \forall \mathbf{w} \in W\}$. We have $\dim W + \dim W^\perp = \dim V$.

Proposition. Suppose that we have a set of maps $f_i : V \to \mathbb{R}$, and points $v_i \in V$, where $1 \leq i \leq n$. If $f_i(v_i) \neq 0$ and $f_i(v_j) = 0$ whenever $i \neq j$, then $f_1, f_2, \ldots, f_n$ are linearly independent.

Definition (Transpose and Symmetric Matrix)**.** For a matrix M, define the *transpose* M^T of M by $M^T_{ij} = M_{ji}$. If $M^T = M$, then M is *symmetric.*

Definition (Determinant)**.** Let M be a $n \times n$ matrix over $\mathbb{F}$. Its *determinant* is

$$\det M = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^{n} M_{i\sigma(i)}.$$

Problems

1. (CWMO 2002) Let $A_1, A_2, \ldots, A_{n+1}$ be non-empty subsets of $[n]$. Prove that there exists nonempty disjoint subsets $I, J \subset [n+1]$ such that

$$\bigcup_{k \in I} A_k = \bigcup_{k \in J} A_k$$

.

2. There are $2n$ people at a party. Each person has an even number of friends at the party (friendship is mutual). Prove that there are two people who have an even number of common friends at the party.

3. (Iran 2006) Let A be a collection of vectors from $\mathbb{Z}_3^n$, with the property that for any two distinct vectors $a, b \in A$, there is some coordinate i such that $b_i = a_i + 1$, where addition is defined modulo 3. Prove that $|A| \leq 2^n$.

4. (IMO 1998 Shortlist C5) In a contest, there are m candidates and n judges, where $n \geq 3$ is an odd integer. Each candidate is evaluated by each judge as either pass or fail. Suppose that each pair of judges agrees on at most k candidates. Prove that

$$\frac{k}{m} \geq \frac{n-1}{2n}.$$

5. (St. Petersburg) Students in a school go for ice cream in groups of at least 2. After $k > 1$ groups have gone, every 2 students have gone together exactly once. Prove the the number of students in the school is at most k.

6. Let $a_1, a_2, \ldots, a_{2n+1} \in \mathbb{R}$, such that for any $1 \leq i \leq 2n+1$, we can remove a_i, and separate the remaining $2n$ numbers into two groups of n numbers with equal sums. Show that $a_1 = a_2 = \cdots = a_{2n+1}$.

7. For even $n \in \mathbb{Z}$, let $S_1, \ldots, S_n$ be subsets of $[n]$ with even size. Prove that there exist some $i \neq j$ such that $|S_i \cap S_j|$ is even.

8. (Iran TST 1996, Germany TST 2004) Let G be a finite simple graph, and there is a light bulb at each vertex of G. Initially, all the lights the off. Each step we are allowed to choose a vertex and toggle the light at that vertex as well as its neighbours'. Show that we can get all the lights to be on at the same time.

9. (Moldova TST 2005) Does there exist a configuration of 22 distinct circles and 22 distinct points on the plane, such that every circle contains at least 7 points and every point belongs at least to 7 circles?

10. (Crux 3037) There are 2021 senators in a senate. Each senator has enemies within the senate. Prove that there is a non-empty subset K of senators such that for every senator in the senate, the number of enemies of that senator in the set K is even.

11. (Fisher) Let $A_1, A_2, \ldots, A_r$ be a collection of distinct subsets of $[n]$, such that for all $i \neq j, |A_i \cap A_j| = t$, where $1 \leq t \leq n$ is some fixed integer. Prove that $r \leq n$.

12. Let $S \subset \mathcal{P}([n])$ such that for all $A \in S$, $|A|$ is odd, and for all distinct $A, B \in S$, $|A \cap B|$ is even. Prove that $|S| \leq n$.

13. Let $S \subset \mathcal{P}([n])$ such that for all $A \in S$, $|A|$ is even, and for all distinct $A, B \in S$, $|A \cap B|$ is odd. Prove that $|S| \leq n + 1$.

14. Let $S \subset \mathcal{P}([n])$ such that for all $A \in S$, $|A|$ is even, and for all distinct $A, B \in S$, $|A \cap B|$ is even. Prove that $|S| \leq 2^{\lfloor \frac{n}{2} \rfloor}$.

15. Let L be a set of s non-negative integers, and let $S \subset [n]^{(r)}$ such that for distinct $A, B \in S, |A \cap B| \in L$. Show that $|S| \leq \displaystyle\sum_{i=0}^{s} \binom{n}{i}$.

16. (Frankl-Wilson) Let p be a prime, $S \subset \mathcal{P}([n])$, and L be a set of s integers. If for all distinct $A, B \in S$, $|A| \notin L \pmod{p}$ and $|A \cap B| \in L \pmod{p}$. Prove that $|S| \leq \displaystyle\sum_{i=0}^{s} \binom{n}{i}$.

17. Let p be a prime and let $S \subset [4p-1]^{(2p-1)}$ be a set system such that for distinct $A, B \in S$, $|A \cap B| \neq p - 1$. Show that $|S| \leq \dfrac{3}{2}\dbinom{4p-1}{p-1}$.

18. (Ray-Chaudhuri-Wilson) Let $0 < k \leq n$ be integers, and let $S \subset \mathcal{P}([n])$, such that for all $A \in S$, $|A| = k$. If L is a set of s positive integers such that for all distinct $A, B \in S, |A \cap B| \in L$. Prove that $|S| \leq \dbinom{n}{s}$.